

Demo: A Battery-Free Sensor Seed for Large-Scale Aerial Deployment in Environmental Monitoring

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Abstract

Large-scale environmental monitoring requires sensing platforms that are ultra-low-cost, lightweight, and easy to deploy over wide outdoor areas. Although existing work has increasingly miniaturized sensor nodes, practical large-scale deployment remains a key bottleneck. In this demo, we present *Sensor Seed*, a bio-inspired aerially deployable sensing platform derived from the maple samara. Fabricated through a monolithic flexible printed circuit process, the proposed design integrates communication and solar-energy-harvesting modules in a compact 300 mg form factor with a unit cost as low as \$5. By co-designing the aerodynamic profile and center of gravity, the sensor seed achieves stable autorotational descent, a solar-exposure landing attitude in over 94% of deployments, and supports communication payloads up to 900 mg while maintaining a terminal velocity below 2.5 m/s. The integrated energy subsystem reaches a 2 V operating voltage within 5 s and 75% state of charge within 100 s under direct solar irradiance. Our demonstration showcases manual and UAV-assisted aerial deployment, stable landing, and rapid post-deployment energy recovery for scalable environmental sensing.

CCS Concepts

• Computer systems organization → Sensors and actuators.

Keywords

Battery-less, Environmental Monitoring, Low-cost Sensor

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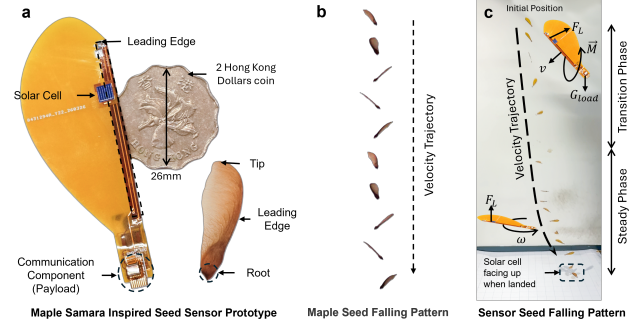


Figure 1: Design and passive aerial behavior of the proposed *Sensor Seed*. (a) Maple-samara-inspired physical prototype with an aerodynamic wing, reinforced leading edge, payload-bearing seed body, and integrated solar cell, shown with a natural maple seed and coin for scale. (b) Falling pattern of a natural maple seed. (c) The wing-leading-edge profile generates lift F_L and aerodynamic moment M , driving stable autorotation and a solar-cell-up landing attitude after aerial deployment.

1 Introduction

Environmental monitoring applications require sensing platforms that can operate for long durations while remaining ultra-low-cost, and easy to deploy at large scale. While existing research has increasingly focused on reducing the footprint and cost of sensor nodes to improve scalability, the practical challenge of efficient large-scale deployment remains largely overlooked [4]. This deployment bottleneck is particularly critical in wide-area agricultural and ecological scenarios, where manual installation can dominate system cost and complexity [1].

To bridge this gap, we present the design and implementation of a bio-inspired, ultra-lightweight sensing seed inspired by the maple samara. The proposed platform combines low-cost fabrication, passive aerial deployment, and solar-powered operation in a compact 300 mg form factor with a unit cost as low as \$5. Through joint optimization of aerodynamic structure and mass distribution, the sensing seed achieves stable autorotational descent, a solar-favorable landing attitude in 94% of deployments, and a low terminal velocity below 2.5 m/s, enabling prompt post-deployment energy harvesting while reducing the need for redundant node deployment.



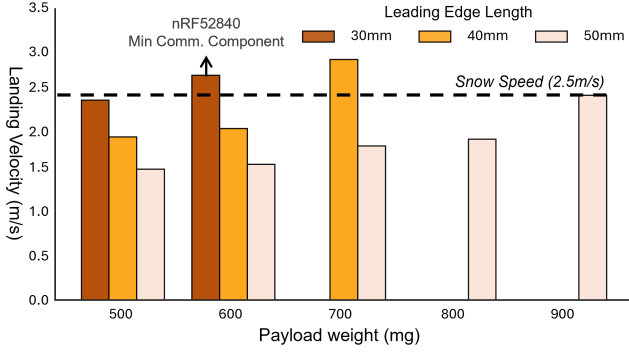


Figure 2: Measured descent velocity under different payload weights and leading-edge lengths.

2 System Architecture

2.1 Maple Inspired Sensor Platform

Fig. 1 shows the biomimetic structural design and passive aerial behavior of the proposed *Sensor Seed*, inspired by the maple samara. As shown in Fig. 1a, the prototype comprises an aerodynamic wing, a reinforced leading edge that provides structural rigidity and promotes leading-edge vortex (LEV) formation, and a payload-bearing seed body integrating the communication and energy harvesting modules. The solar cell is positioned near the wing root, while the electronic payload is concentrated at the seed body to tune the center of gravity (CoG) and emulate the asymmetric mass distribution of its biological counterpart [3]. The structure is realized through a monolithic Flexible Printed Circuit (FPC) process using a 75 μm polyimide (PI) wing and a 0.2 mm reinforced PI leading edge.

Fig. 1b illustrates the characteristic falling pattern of a natural maple seed, which serves as the target passive descent behavior for our design. Accordingly, the proposed sensor seed is engineered to reproduce a similar steady-state autorotational descent and controlled landing process, as illustrated in Fig. 1c. When released in a leading-edge-down posture, the wing-leading-edge profile generates lift F_L and an aerodynamic moment M , driving rotational stabilization and biasing the final landing posture toward a solar-cell-up configuration.

By calibrating the CoG of the integrated components, we regulate the competing torques induced by aerodynamic lift F_L and gravity G_{load} during descent, enabling solar-exposure landing attitude for sustained energy harvesting [2]. When freely released from the initial posture shown in Fig. 1c, the sensor seed achieves a solar-exposure landing attitude in over 94% of deployments across 50 trials. Experimental results further show that the sensor seed supports communication payloads of up to 900 mg, including practical wireless platforms such as the nRF 52840 (~600 mg minimum communication subsystem), while maintaining a terminal velocity below 2.5 m/s from a 3 m free release, reducing landing impact and improving deployment robustness (Fig. 2).

2.2 Energy Harvest Platform

Following solar-favorable landing, we implement an energy subsystem that exploits the seed's upward-facing solar orientation

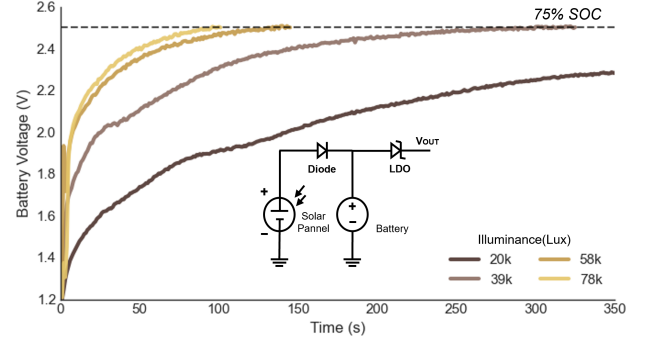


Figure 3: Charging performance of the proposed energy subsystem under different illuminance levels.

for sustained post-deployment operation. To avoid the frequent cold-start interruptions common in capacitor-buffered designs, we integrate a Gallium Arsenide solar panel with a high-density micro-battery (PWY0150S-ITEN, 3.5 mm $\times$ 5.1 mm, 73 mg). This design provides an energy capacity comparable to a 0.4 F supercapacitor while achieving a superior balance between energy density and aerodynamic payload. As shown in Fig. 3, the system reaches a 2 V operating voltage within 5 s and 75% state of charge (SOC) within 100 s under direct solar irradiance (78,000 lux). The stored energy further acts as a buffer against temporary solar occlusion, sustaining up to 5,500 s of active data processing or the transmission of 8 consecutive BLE data packets in experiment, thereby ensuring mission continuity in dynamic outdoor environments.

3 Demonstration

To validate the practical feasibility of the proposed system, we will conduct field demonstrations including both manual release and UAV-assisted dispersal to emulate high-altitude aerial deployment. These demonstrations will highlight the stable autorotational descent, solar-favorable landing behavior, and low-impact ground contact of the proposed sensor seed in realistic outdoor environments. In addition, we will showcase the rapid energy recovery capability of the integrated energy subsystem under controlled solar-emulator illumination, demonstrating the system's ability to quickly reach an operational power state and sustain sensing and communication under varying lighting conditions.

Demo Link: <https://youtu.be/LXlhCeMjxoE>

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